INCH-POUND

MIL-S-19500/315E
10 March 1992
SUPERSEDING
MIL-S-19500/315D
12 October 1979

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N2880, 2N3749, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors for use in high-speed power-switching applications. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. Type 2N2880, see figure 1.
Type 2N3749, see figure 2.

1.3 Maximum ratings.

P_T 1/ T_A = 25°C	P_T 2/ T_C = 100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{STG} and T_{OP}
W	W	V dc	V dc	V dc	A dc	A dc	°C
2	30	110	80	8	5	0.5	-65 to +200

1/ Derate linearly 11.4 mW/°C for T_A > 25°C.
2/ Derate linearly 300 mW/°C for T_C > 100°C.

1.4 Primary electrical characteristics at T_C = 25°C.

Limits	h_{FE3} 1/ V_{CE} = 5 V dc, I_C = 5 A dc	h_{FE2} 1/ V_{CE} = 2 V dc, I_C = 1 A dc	$\|h_{fe}\|$ V_{CE} = 10 V dc, I_C = 1 A dc, f = 10 MHz	$V_{BE(sat)}$ 1/ I_C = 1 A dc, I_B = 100 mA dc	$V_{CE(sat)}$ 1/ I_C = 1 A dc, I_B = 100 mA dc	C_{obo} V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	$R_{\theta JC}$
				V dc	V dc	pF	°C/W
Min	15	40	3	---	---	---	---
Max	---	120	12	1.2	0.25	150	3.33

1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, Attn: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5280, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

$R_{(ISO)}$ - - - - - - - - - - - - - Insulation resistance.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 and 2 herein. Current density of internal conductors shall be as specified in MIL-S-19500.

3.3.1 Lead material and finish. Lead material and finish shall be in accordance with MIL-S-19500 and MIL-STD-750. Where a choice of lead material or finish is desired, it shall be specified in the contract or order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500 except at the option of the manufacturer, the country of origin marking may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

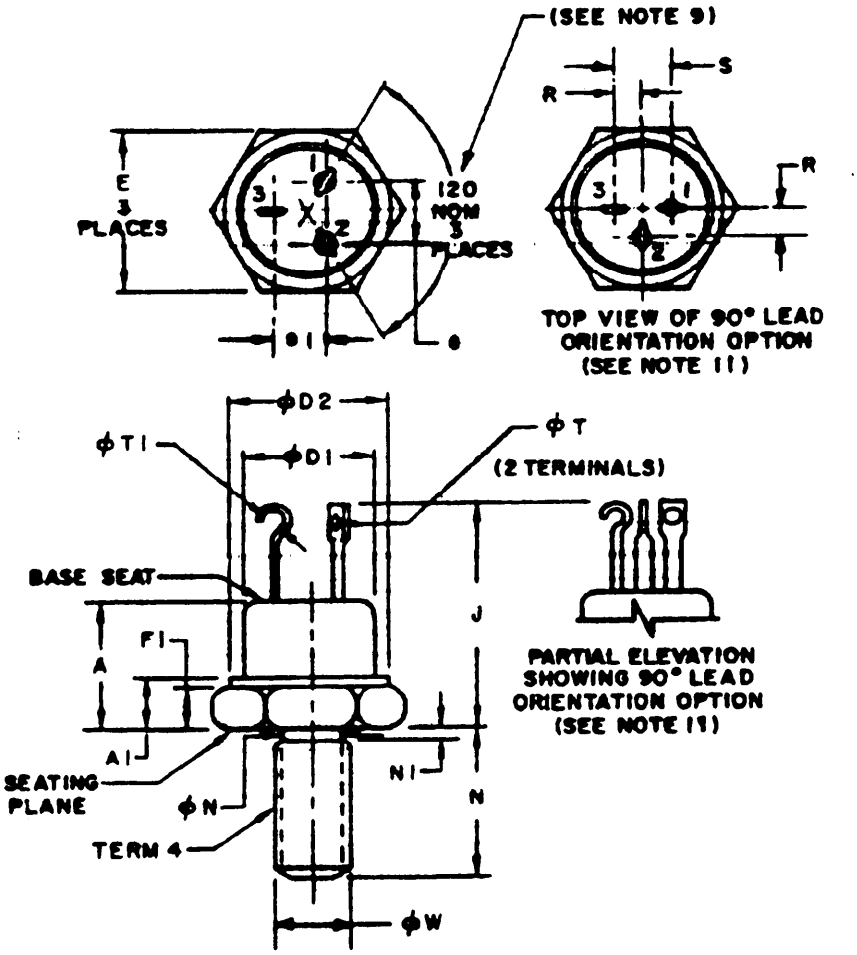


FIGURE 1. Physical dimensions of transistor type 2N2880.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.320	.468	8.13	11.89	
A_1	---	.250	---	6.35	
ϕD_1	.318	.380	8.08	9.65	
ϕD_2	.380	.437	9.65	11.10	
E	.423	.438	10.74	11.13	
e	.125	.165	3.18	4.19	5,8,9
e_1	.110	.145	2.79	3.68	5,8
F	.090	.150	2.29	3.81	
J	.570	.763	14.48	19.38	4
ϕN	.155	.189	3.94	4.80	
N	.400	.455	10.16	11.56	
N_1	---	.078	---	1.98	10
ϕT	.040	.065	1.02	1.65	
ϕT_1	.040	.070	1.02	1.78	
ϕW	.190-32UNF-2A				4
R	.090	.110	2.29	2.79	5,8,9
S	.185	.215	4.70	5.46	5,8,9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Collector shall be electrically connected to the case. This terminal may be flattened and pierced only when the 90° option is used.
4. ϕW is the outer diameter of coated threads. Reference: Screw thread standards for Federal Service Handbook H28, part I.
5. The orientation of the terminals in relation to the hex flats is not controlled.
6. All three terminals.
7. The case temperature may be measured anywhere on the seating plane within .125 (3.18 mm) of the stud.
8. Terminal spacing measured at the base seat only.
9. Dimensions e, e_1, R, and S are measured from the center line of terminals.
10. Maximum unthreaded dimension.
11. This dimension applies to the location of the center line of the terminals.
12. A 90° angle lead orientation as shown may be used at the option of the manufacturer. All dimensions of the basic outline except e, e_1, and the 120° lead angle apply to this option.
13. Terminal -1, Emitter; Terminal -2, Base; Terminal -3, Collector.
14. A slight chamfer or undercut on one or both ends of the hexagonal is optional.

FIGURE 1. Physical dimensions of transistor type 2N2880 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

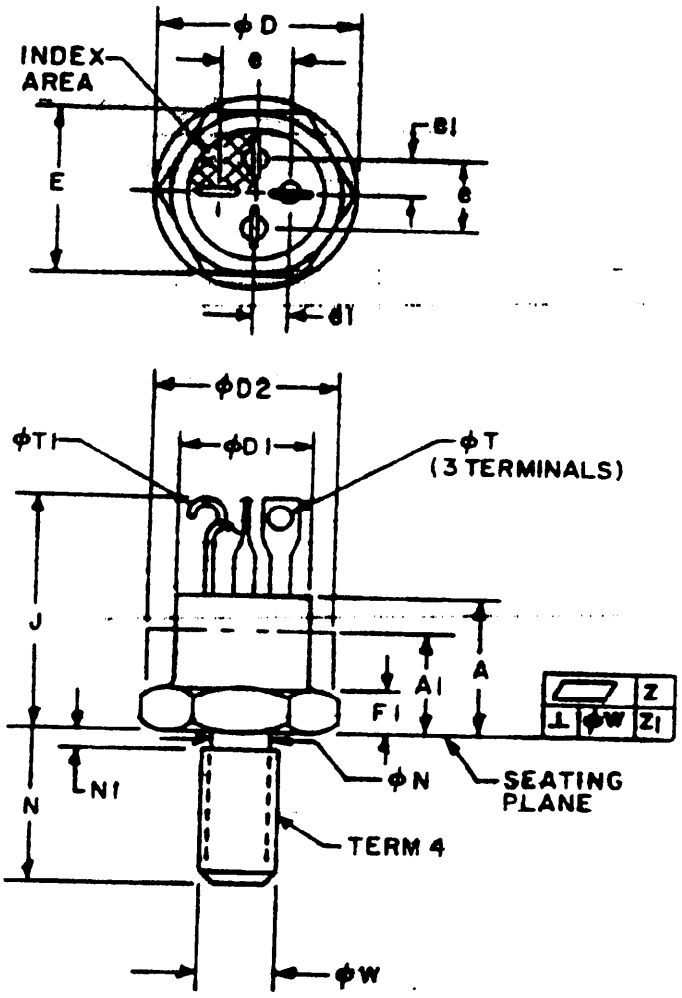


FIGURE 2. Physical dimensions of transistor type 2N3749.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.320	.458	8.13	11.89	
A_1	---	.250	---	6.35	5
ϕO	---	.506	---	12.83	
ϕO_1	.318	.380	8.08	9.65	
ϕO_2	.380	.437	9.65	11.10	5
E	.423	.438	10.74	11.13	
e	.180	.215	4.57	5.46	7
e_1	.080	.110	2.03	2.79	7
F_1	.090	.150	2.29	3.81	4,8
J	.570	.763	14.48	19.38	
M	.155	.189	3.94	4.80	
N	.400	.455	10.16	11.56	
N_1	---	.078	---	1.98	9
ϕT	.040	.065	1.02	1.65	
ϕT_1	.040	.070	1.02	1.78	6
ϕW	.190-32UNF-2A				10
Z	---	.002	---	.05	
Z_1	---	.006	---	.15	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal 1 - Emitter; Terminal 2 - Base; Terminal 3 - Collector; Terminal 4 - Case.
4. Chamfer or undercut on one or both ends of hexagonal portion is optional.
5. The outline contour with the exception of the hexagon is optional within cylinder defined by ϕO_2 and A_1.
6. Terminal 4 can be flattened and pierced or hook type. A visual index is required when the flattened and pierced tab terminal contour (identical to the adjacent terminals) option is used.
7. Angular orientation of terminals with respect to hexagon is optional.
8. F_1 dimension does not include sealing flanges.
9. N_1 is the length of incomplete or undercut threads.
10. ϕW is the outer diameter of coated threads. Reference: Screw thread standards for Federal Service Handbook H28, part I.

FIGURE 2. Physical dimensions of transistor type 2N3749 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CBO1} and h_{FE2}	I_{CBO1}
11	I_{CBO1} and h_{FE2}; ΔI_{CBO1} = 100% of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = +15%, -10%	I_{CBO1} and h_{FE2} ΔI_{CBO1} = 100% of initial value or 100 nA dc, whichever is greater.
12	See 4.3.1	See 4.3.1
13(a)	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = +15%, -10%	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 100 nA dc, whichever is greater; Δh_{FE2} = +20%, -10%
13(b) (2N3749 only)	MIL-STD-750, method 1016 insulation resistance test condition B (short connector, emitter, and base terminals together), R_{ISO} = 10^9 ohms (min)	MIL-STD-750, method 1016 insulation resistance test condition B (short connector, emitter, and base terminals together), R_{ISO} = 10^9 ohms (min)

4.3.1 Power burn-in conditions. Power burn-in conditions (all levels) are as follows:

T_J = 187.5 ±12.5°C $T_A \leq 100°C$ V_{CE} = 25 ±5 V dc

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. (Endpoint electrical measurements shall be in accordance with the applicable steps of table IV herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table IV herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2 2/						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	110	---	V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	80	---	V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	8	---	V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 80 V dc	I_{CBO1}	---	0.2	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 110 V dc; V_{BE} = -0.5 V dc	I_{CEX1}	---	1.0	µA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 60 V dc	I_{CEO}	---	20	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO}	---	0.2	µA dc
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 50 mA dc	h_{FE1}	40	120	---
Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE2}	40	120	---
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE3}	15	---	---
Collector to emitter voltage (saturated)	3071	I_C = 1.0 A dc; I_B = 0.1 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$	---	0.25	V dc
Collector to emitter voltage (saturated)	3071	Test condition A; I_C = 5.0 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$	---	1.5	V dc

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
2/ Subgroup 2 - Continued						
Base to emitter voltage (non-saturated)	3066	Test condition B; V_{CE} = 2 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	V_{BE}	---	1.2	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 1 A dc; I_B = 0.1 A dc; pulsed (see 4.5.1)	$V_{BE(sat)}$	---	1.2	V dc
Subgroup 3						
High-temperature operation:		T_A = 150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 80 V dc; V_{BE} = 0.5 V dc	I_{CEX2}	---	50	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}	---	10	µA dc
Low-temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 A dc	h_{FE4}	15	---	---
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 50 mA dc; f = 1 kHz	h_{fe}	40	140	---
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1 A dc; f = 10 MHz	$\|h_{fe}\|$	3	12	---
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 ≤ f ≤ 1 MHz	C_{obo}	---	150	pF
Switching parameters:						
Pulse delay time	---	See figure 3	t_d	---	60	ns
Pulse rise time	---	See figure 3	t_r	---	300	ns
Pulse storage time	---	See figure 3	t_s	---	1.7	µs
Pulse fall time	---	See figure 3	t_f	---	300	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (D.C.)	3051	T_C = +100°C; Power application time = 10s (see figure 4)				
Test 1		V_{CE} = 80 V dc I_C = 80 mA dc	---	---	---	---
Test 2		V_{CE} = 20 V dc I_C = 1.5 A dc	---	---	---	---
Safe operating area (clamped switching)		See figures 6 and 7 (clamped inductive load) T_A = +25°C; I_B = 0.5 A dc; I_C = 5 A dc; V_{CC} = 15 V dc				
Endpoint electrical measurements		See table IV, steps 1, 3, 7, and 9				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.
2/ For JANS level, all devices required by the specified LTPD shall be subjected to subgroups 2, 3, and 4 combined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE IIa. Group B inspection for JANS devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figures 1 and 2
Subgroup 2		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 3		
Thermal shock (temperature cycling)	1051	
Hermetic seal a. Fine leak b. Gross leak	1071	
Electrical measurements		See table IV, steps 1, 3, 5, 6, and 8
Decap internal visual design verification	2075	
Bond strength	2037	Test condition A. All internal wires for each device shall be pulled seperately.
Die shear	2017	
Subgroup 4		
Intermittent operation life	1037	V_{CE} = 5 V dc; P_T = 2 W T_A = 25 ±3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink nor forced air cooling on the devices shall be permitted.
Electrical measurements		See table IV, steps 1, 3, 5, 6, 8, 9, 11, and 12
Subgroup 5		
Accelerated steady state operation life	1027	V_{CE} = 5 V dc; T_A = 125°C ±25°C for 96 hours. P_T 2W at T_A = 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°. Marking legibility requirements shall not apply.

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

See footnote at end of table.

TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 5 - Continued		
Electrical measurements		See table IV, steps 1, 3, 5, 6, 8, 9, 10, 11, and 12
Bond strength (Al - Au die interconnects only)	2037	Test condition A
Subgroup 6		
Thermal resistance	3131	See 4.5.2
Subgroup 7		
Safe operating area (unclamped inductive)	3053	Load condition C; 22 devices; c = 0; (unclamped inductive load) (see figure 5) T_A = 25°C; duty cycle ≤ 10%; R_S = 0.1Ω; T_p = 640 µs
Test 1		R_{BB1} = 39Ω; V_{BB1} = 20 V dc; R_{BB2} = ∞; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 5 A dc; L = 1 mH; (0.5Ω, 5 A dc) (tower #7870 or equivalent)
Test 2		T_p = 2.88 mS; R_{BB1} = 120Ω; V_{BB1} = 20 V dc; R_{BB2} = ∞; V_{BB} = 0; V_{CC} = 15 V dc; I_C = 1.6 A dc; L = 10 mH; (0.11Ω, 12.5 A dc) (stancor c-2688 or equivalent)
Safe operating area (clamped switching)		See figures 6 and 7 (clamped inductive load) T_A = 25°C; I_B = 0.5 A dc; I_C = 5 A dc; V_{CC} = 55 V dc (destructive test)
Electrical measurements		See table IV, steps 1, 3, 7, and 9

1/ For sampling plan, if not listed, see MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE IIb. Group B inspection for JANTX and JANTXV devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	
Hermetic seal a. fine leak b. gross leak	1071	
Electrical measurements		See table IV, steps 1, 3, 5, and 9.
Subgroup 3		
Steady state operation life	1027	T_J = 187.5 ±12.5°C; $T_A \leq$ 100°C; V_{CE} = 25 ±5 V dc
Electrical measurements		See table IV, steps 2, 4, 5, 9, and 11.
Subgroup 4		
Decap internal visual design verification	2075	
Bond strength	2037	Test condition A; all internal wires for each device shall be pulled separately.
Subgroup 5		
Thermal resistance	3131	See 4.5.2
Subgroup 6		
High-temperature life (nonoperating)	1032	T_A = +200°C
Electrical measurements		See table IV, steps 1, 4, 5, 9, 10, and 11

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE IIb. Group B inspection for JANTX and JANTXV devices - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 7		
Safe operating area	3053	Load condition C; (unclamped inductive load) (see figure 5); T_A = 25°C; duty cycle ≤ 10%; R_S = 0.1Ω
Test 1		T_p = 640 µs; R_{BB1} = 39Ω; V_{BB1} = 20 V dc; R_{BB2} = ∞; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 5 A dc; L = 1 mH; (0.5Ω; 5 A dc) (tower #7870 or equivalent)
Test 2		T_p = 2.88 ms; R_{BB1} = 120Ω; V_{BB1} = 20 V dc; R_{BB2} = ∞; V_{BB} = 0; V_{CC} = 15 V dc; I_C = 1.6 A dc; L = 10 mH; (0.11Ω; 12.5 A dc) (stancor c-2688 or equivalent)
Safe operating area (clamped switching)		See figures 6 and 7 (clamped inductive load) T_A = 25°C; I_B = 0.5 A dc; I_C = 5 A dc; V_{CC} = 55 V dc; (destructive test)
Electrical measurements		See table IV, steps 1, 3, 7, and 9

1/ For sampling plan, see MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE III. Group C inspection (all quality levels).

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figures 1 and 2
Subgroup 2		
Thermal shock (glass strain)	1056	Test condition B
Terminal strength (terminal torque)	2036	Test condition B1; torque = 6 inch-ounces; t = 15 s; (tubulated leads only)
(stud torque)		Test condition D2; torque = 15 inch-pound; application time = 15 s
(tension)		Test condition A; weight = 7 pounds ±5 ounces; application time = 15 s (tubulated leads only)
Hermetic seal a. Fine leak b. Gross leak	1071	
Moisture resistance	1021	
External visual	2071	
Electrical measurements		See table IV, steps 1, 3, 5, and 6 (JANS) and steps 1, 3, 5, 6, and 9 (JAN, JANTX, and JANTXV)
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	
Constant acceleration	2006	
Electrical measurements		See table IV, steps 5 and 6 (all quality levels)
Subgroup 4		
Salt atmosphere (corrosion)	1041	

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE III. Group C inspection (all quality levels) - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 5		
Not applicable		
Subgroup 6		
Steady state operation life	1026	V_{CE} = 20 V dc; P_D = 30 W at T_C = +100°C
Electrical measurements		See table IV, steps 3, 5, 6, 9, 10, and 11 (JANS) and steps 2, 4, 5, 9, and 11 (JAN, JANTX, and JANTXV)

1/ For sampling plan, see MIL-S-19500.

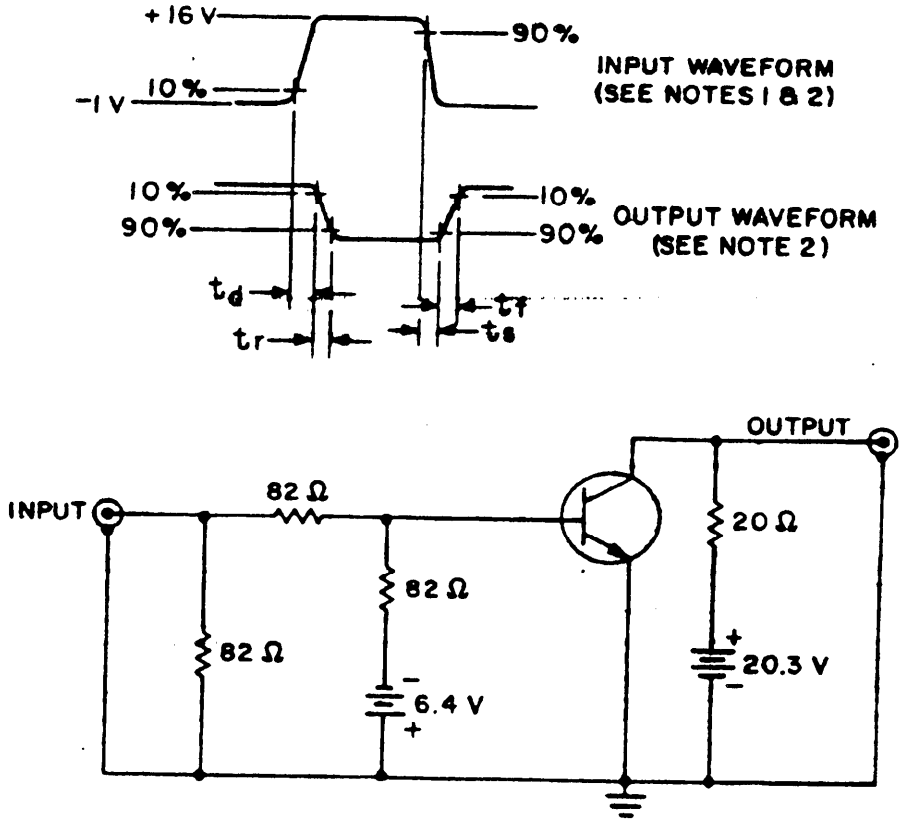
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

TABLE V. Groups A, B, C, and E electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 80 V dc	I_{CBO1}	---	0.2	µA dc
2.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 80 V dc	I_{CBO1}	---	0.4	µA dc
3.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO}	---	0.2	µA dc
4.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO}	---	0.4	µA dc
5.	Collector to emitter voltage (saturated)	3071	I_C = 1.0 A dc; I_B = 0.1 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$	---	0.25	V dc
6.	Base to emitter voltage (saturated)	3066	Test condition A; I_C = 1.0 A dc; I_B = 0.1 A dc	$V_{BE(sat)}$	---	1.2	V dc
7.	Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 0.1 A dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	80	---	V dc
8.	Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE1}	40	120	---
9.	Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE2}	40	120	---
10.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 80 V dc	ΔI_{CBO1} 1/	---	100% of initial value or 100 nA dc, whichever is greater.	
11.	Forward current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	Δh_{FE2} 1/		+20%, -10% change from initial reading.	
12.	Collector to emitter voltage (saturated)	3071	I_C = 1.0 A dc; I_B = 0.1 A dc; pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$ 1/		50 mV dc change from previously measured value.	

1/ Devices which exceed the group A limits for this test shall not be accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.



NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics:

   $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\Omega$, PW = 2 μs, duty cycle $\leq 2\%$.

2. Output waveforms are monitored on an oscilloscope with the following characteristics:

   $t_r \leq 15$ ns, $R_{in} \geq 10$ MΩ, $C_{in} \leq 11.5$ pF.

3. Resistors shall be noninductive types.
4. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 3. Pulse response test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

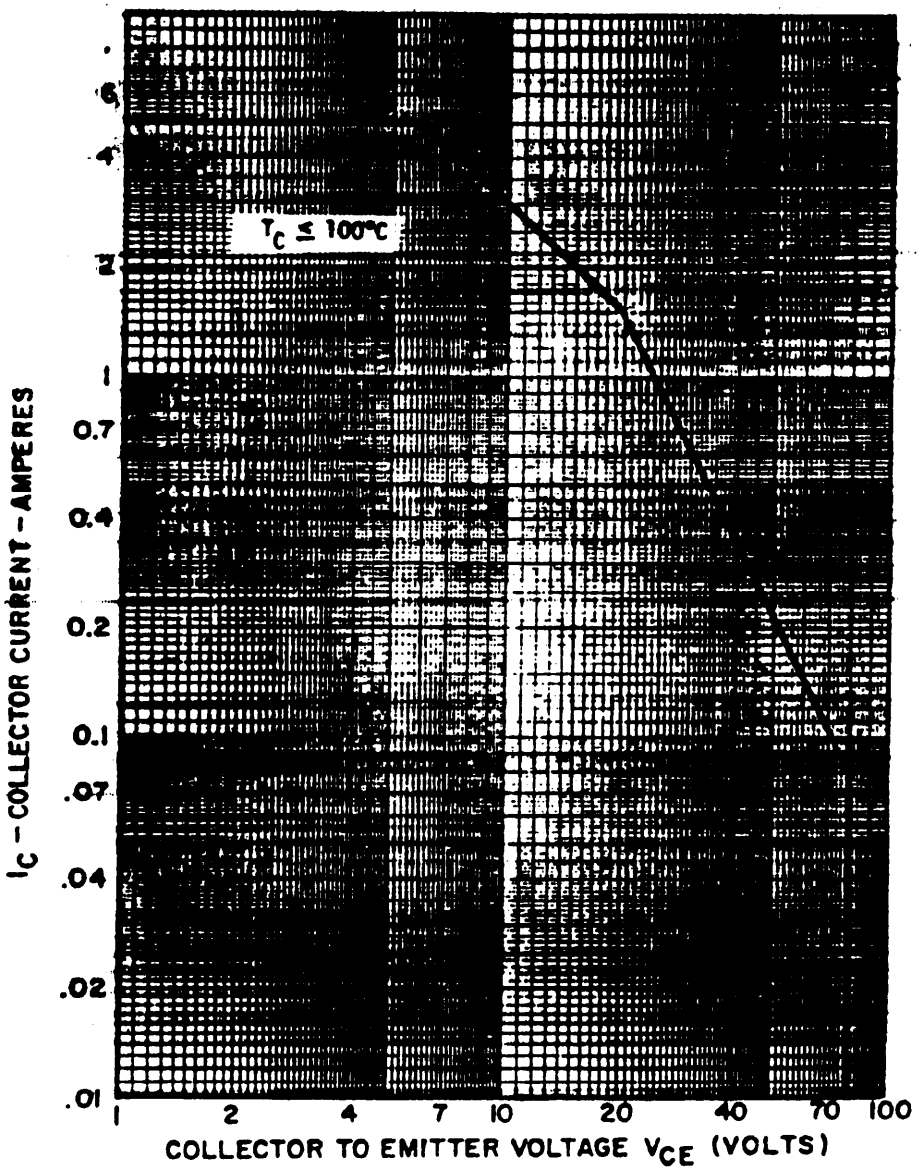


FIGURE 4. Maximum safe operating area graph (dc).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

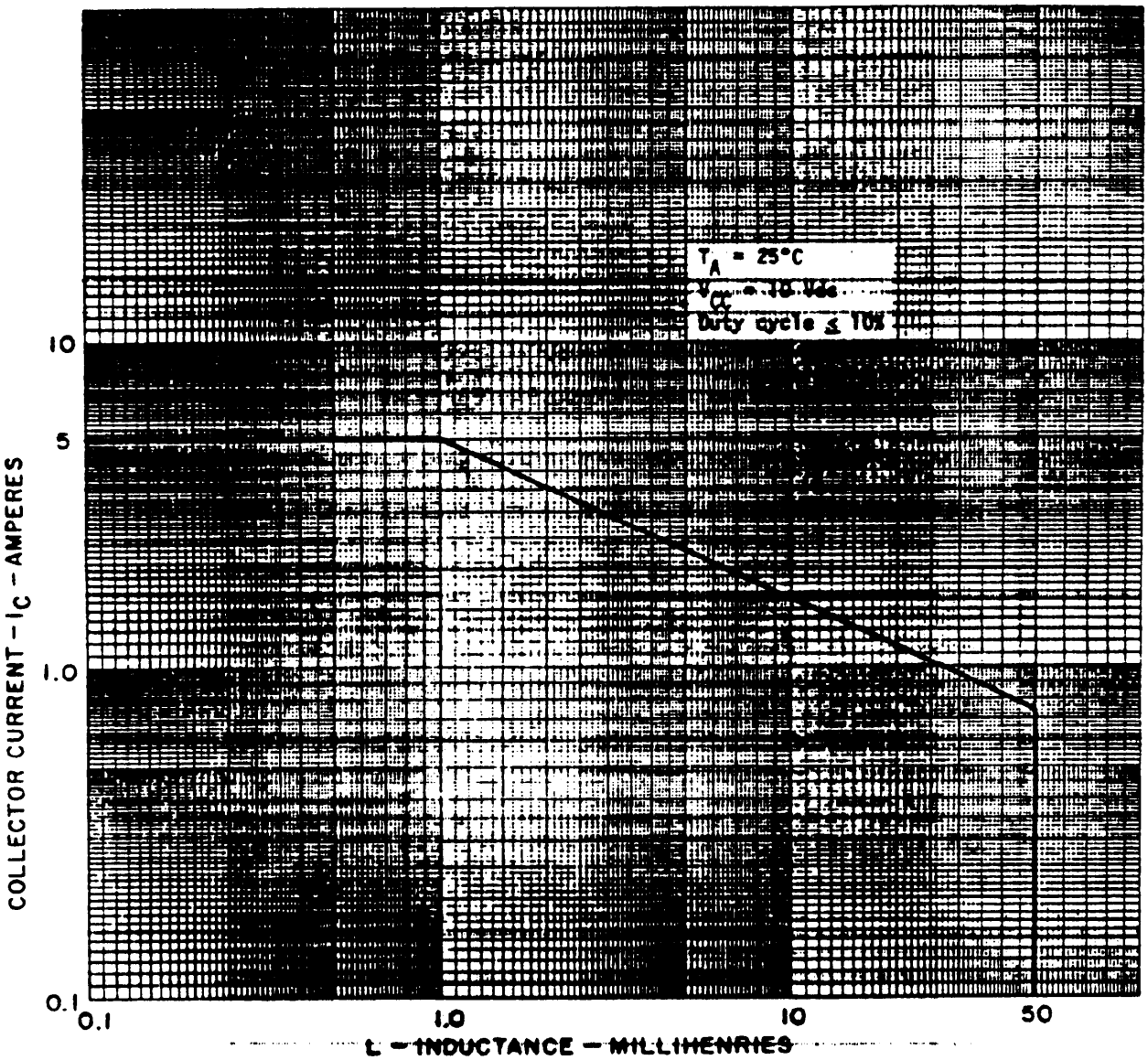
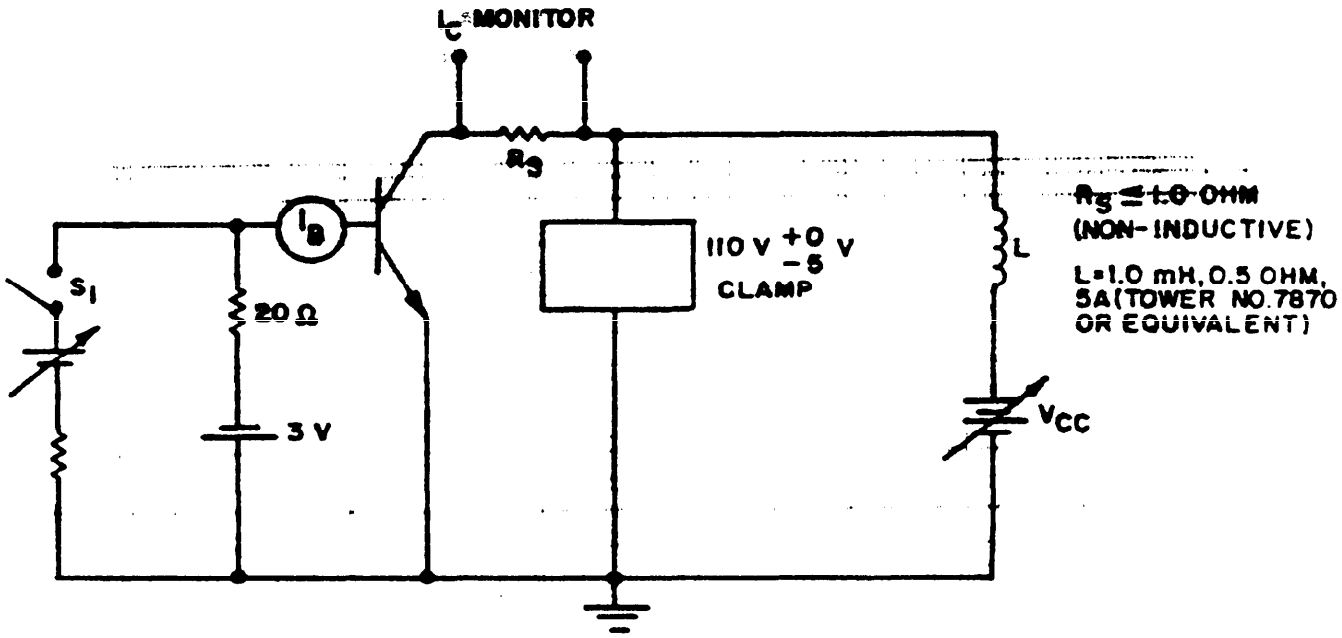


FIGURE 5. Safe operating area for switching between saturation and cutoff - unclamped inductive load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.



Procedure:
1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if clamp voltage is not reached.
3. Perform specified endpoint tests.

FIGURE 6. Clamped inductive sweep test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

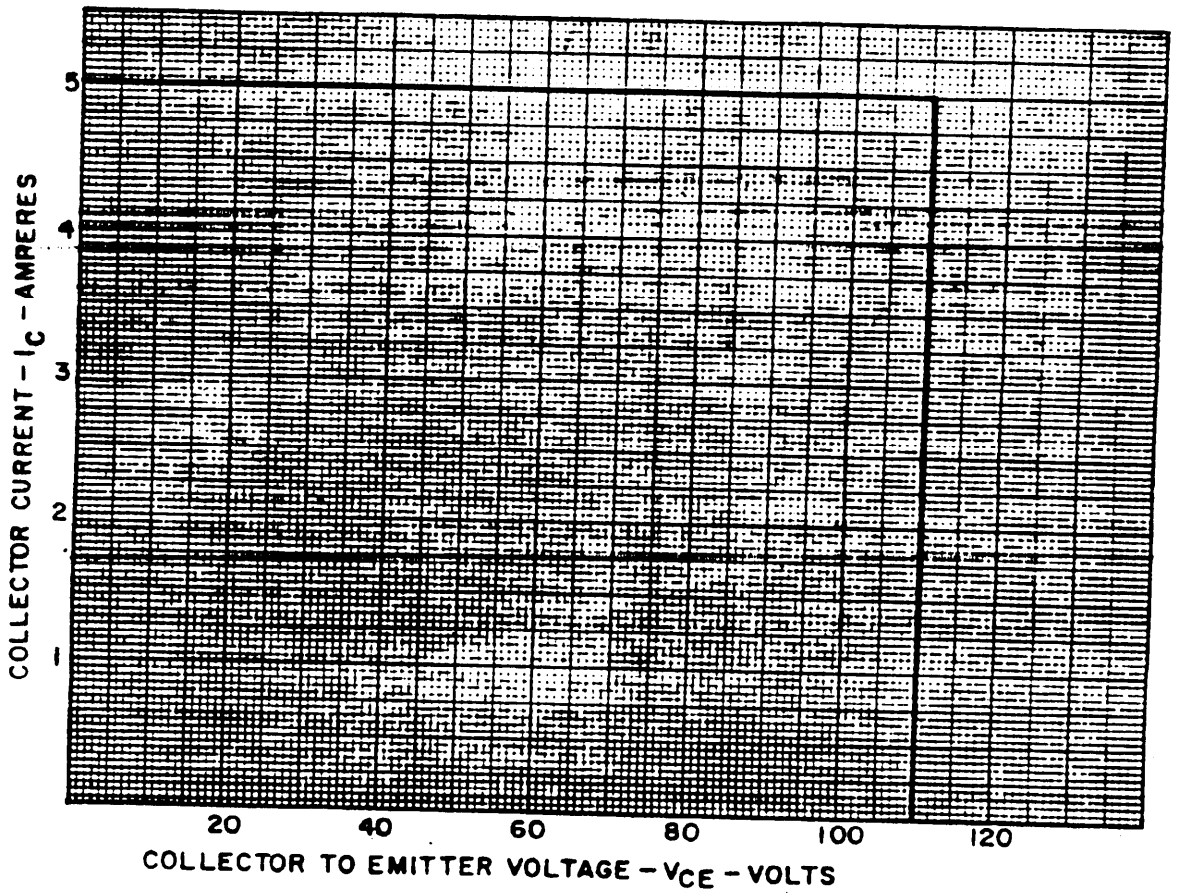


FIGURE 7. Safe operating area for switching between saturation and cutoff - clamped inductive load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be selected with .8 A dc ≤ I_C ≤ 1.2 A dc and recorded before test is started.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be selected with 25°C ≤ t_R 75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit of $R_{\theta JC}$ shall be 3.33°C/W.

4.5.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted at a case temperature (t_C) of 25°C.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Type number and quality level.

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced herein (see 2.1).

d. Lead material and finish as applicable (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES
NASA - LRC, MSF

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH
Air Force - 14

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1270)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:17Z
Check the source to verify that this is the current version before use.